Configuring DTSA-II

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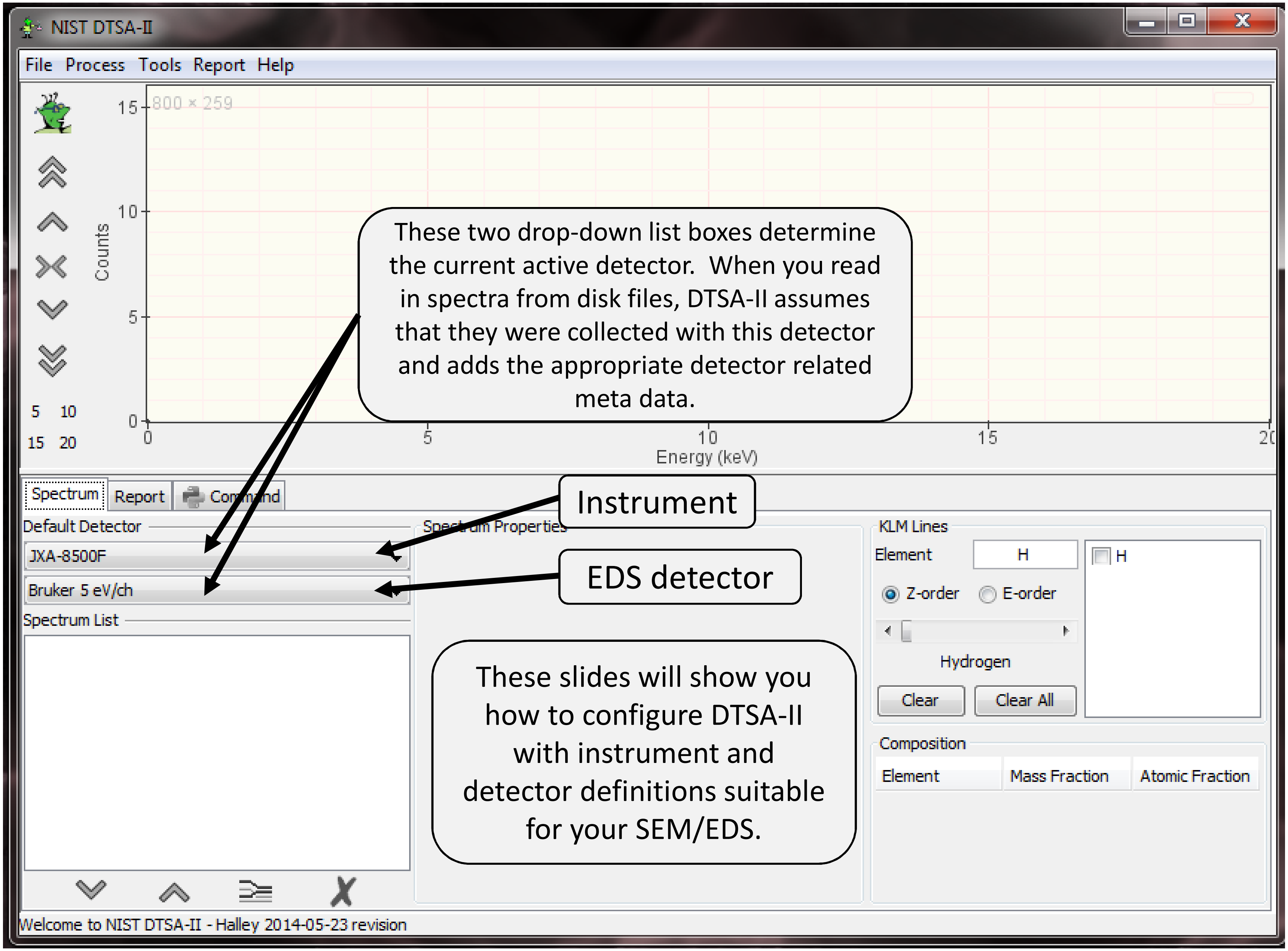
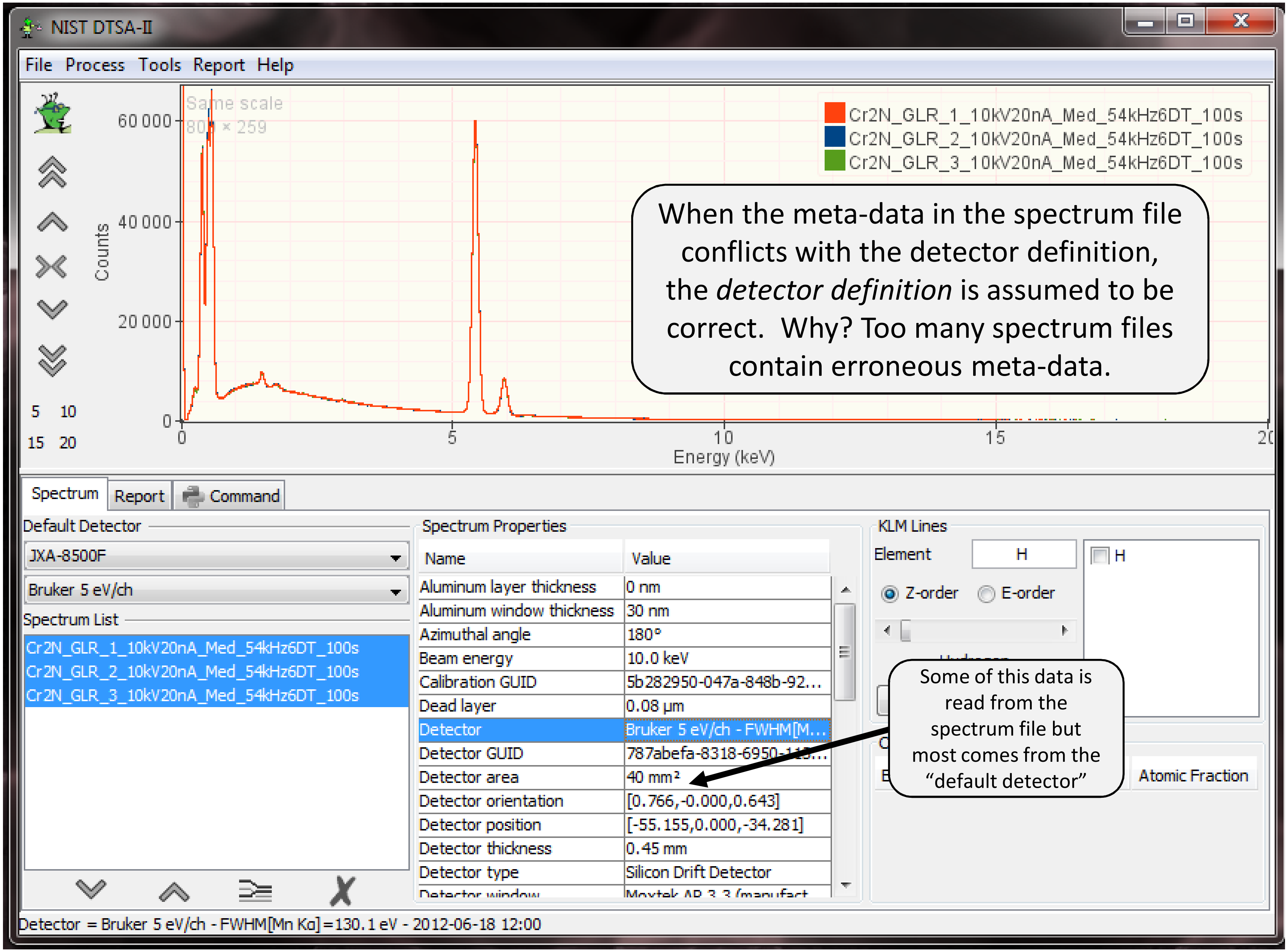
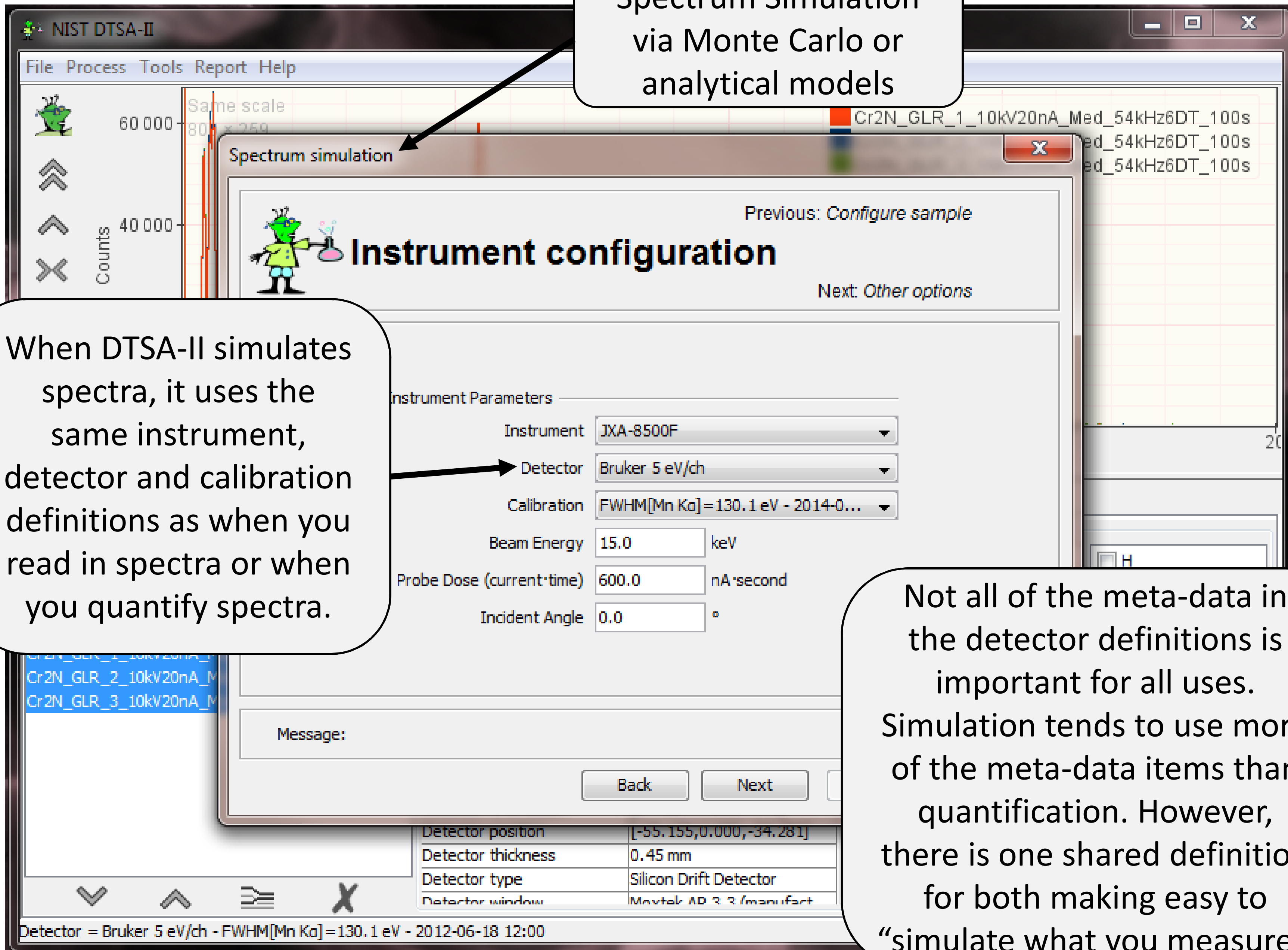


NIST DTSA-II
File Process Tools Report Help
Counts
Energy (keV)
These two drop-down list boxes determine the current active detector. When you read in spectra from disk files, DTSA-II assumes that they were collected with this detector and adds the appropriate detector related meta data.
Spectrum
Report
Command
Default Detector
JXA-8500F
Bruker 5 eV/ch
Spectrum List
Spectrum Properties
Instrument
EDS detector
These slides will show you how to configure DTSA-II with instrument and detector definitions suitable for your SEM/EDS.
KLM Lines
Element
H
Z-order
E-order
Hydrogen
Clear
Clear All
Composition
Element
Mass Fraction
Atomic Fraction
Welcome to NIST DTSA-II - Halley 2014-05-23 revision

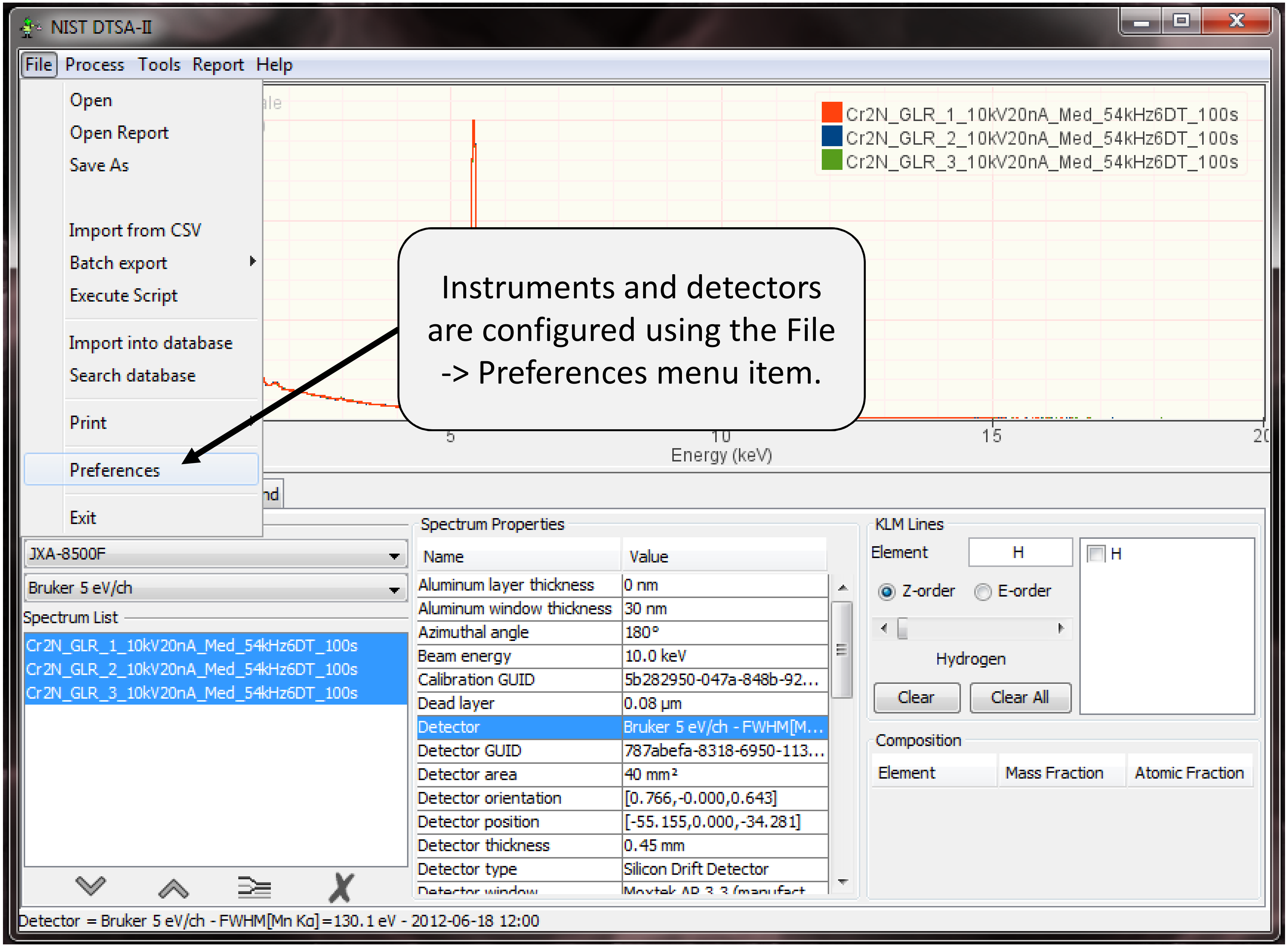

NIST DTSA-II
File Process Tools Report Help
Same scale
Counts
Energy (keV)
Cr2N_GLR_1_10kV20nA_Med_54kHz6DT_100s
Cr2N_GLR_2_10kV20nA_Med_54kHz6DT_100s
Cr2N_GLR_3_10kV20nA_Med_54kHz6DT_100s
When the meta-data in the spectrum file conflicts with the detector definition, the *detector definition* is assumed to be correct. Why? Too many spectrum files contain erroneous meta-data.
Spectrum Report Command
Default Detector
JXA-8500F
Bruker 5 eV/ch
Spectrum List
Spectrum Properties
Name Value
Aluminum layer thickness 0 nm
Aluminum window thickness 30 nm
Azimuthal angle 180°
Beam energy 10.0 keV
Calibration GUID 5b282950-047a-848b-92...
Dead layer 0.08 µm
Detector Bruker 5 eV/ch - FWHM[M...
Detector GUID 787abefa-8318-6950-11...
Detector area 40 mm²
Detector orientation [0.766,-0.000,0.643]
Detector position [-55.155,0.000,-34.281]
Detector thickness 0.45 mm
Detector type Silicon Drift Detector
KLM Lines
Element H
Z-order E-order
Atomic Fraction
Some of this data is read from the spectrum file but most comes from the "default detector"
Detector = Bruker 5 eV/ch - FWHM[Mn Kα]=130.1 eV - 2012-06-18 12:00

Spectrum Simulation via Monte Carlo or analytical models

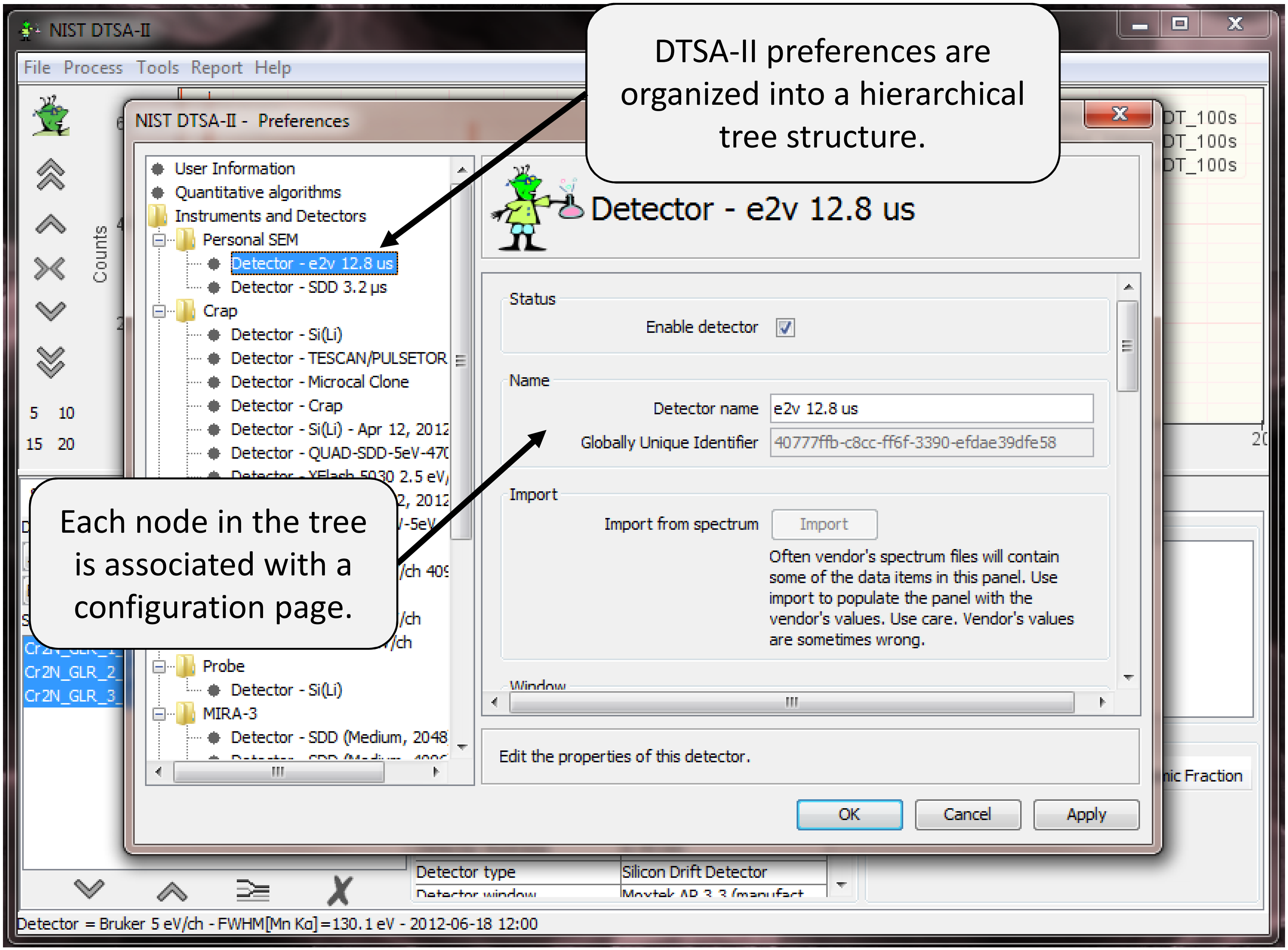


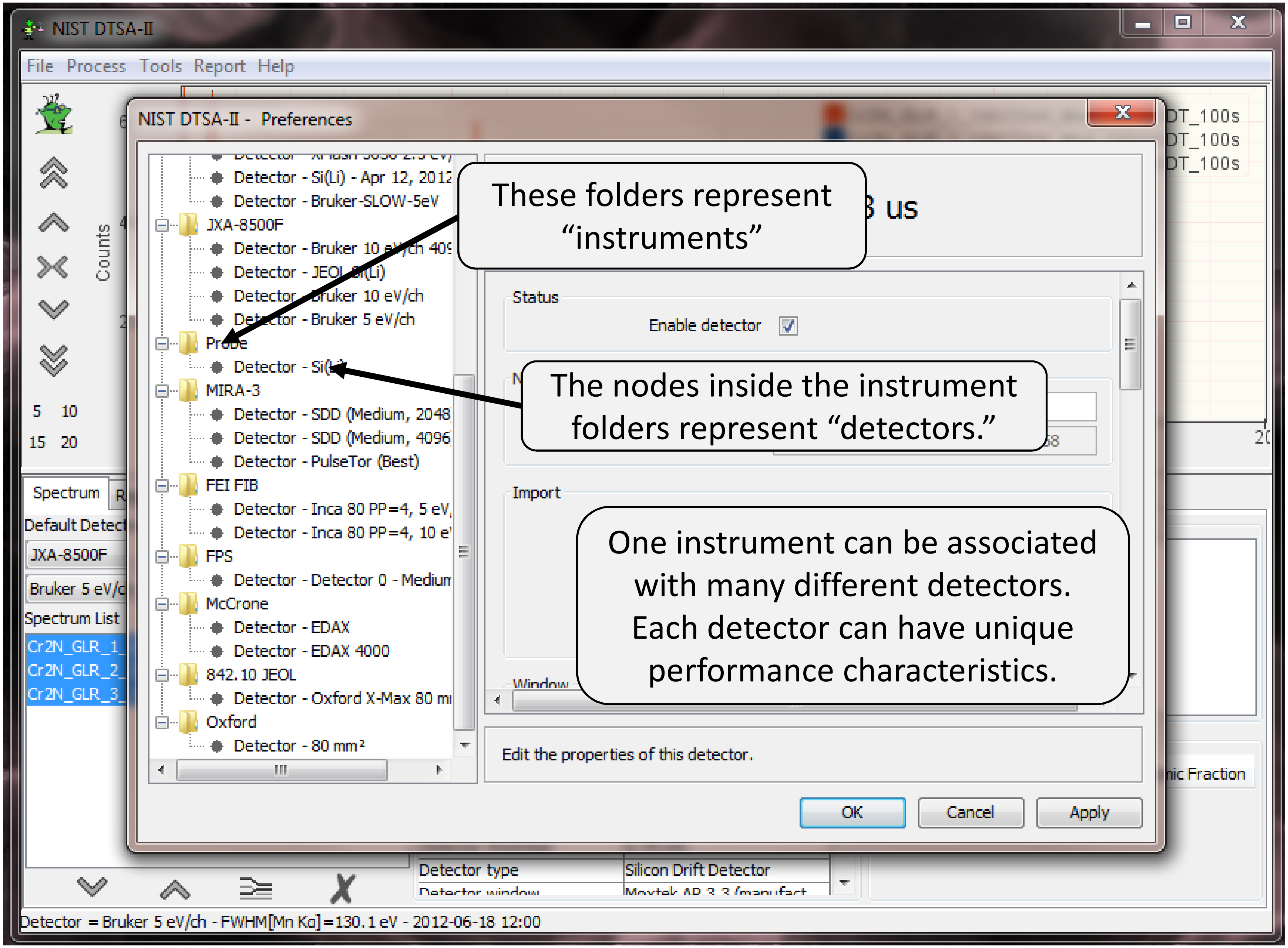
When DTSA-II simulates spectra, it uses the same instrument, detector and calibration definitions as when you read in spectra or when you quantify spectra.

Not all of the meta-data in the detector definitions is important for all uses. Simulation tends to use more of the meta-data items than quantification. However, there is one shared definition for both making easy to "simulate what you measure".



Nicholas W. M. Ritchie, NIST


DTSA-II preferences are organized into a hierarchical tree structure.
Each node in the tree is associated with a configuration page.
NIST DTSA-II
File Process Tools Report Help
NIST DTSA-II - Preferences
User Information
Quantitative algorithms
Instruments and Detectors
Personal SEM
Detector - e2v 12.8 us
Detector - SDD 3.2 µs
Crap
Detector - Si(Li)
Detector - TESCAN/PULSETOR
Detector - Microcal Clone
Detector - Crap
Probe
Detector - Si(Li)
MIRA-3
Detector - e2v 12.8 us
Status
Enable detector
Name
Detector name
e2v 12.8 us
Globally Unique Identifier
40777ffb-c8cc-ff6f-3390-efdae39dfe58
Import
Import from spectrum
Import
Often vendor's spectrum files will contain some of the data items in this panel. Use import to populate the panel with the vendor's values. Use care. Vendor's values are sometimes wrong.
Edit the properties of this detector.
OK
Cancel
Apply
Detector type
Silicon Drift Detector
Detector = Bruker 5 eV/ch - FWHM[Mn Kα]=130.1 eV - 2012-06-18 12:00


NIST DTSA-II
File Process Tools Report Help
NIST DTSA-II - Preferences
These folders represent “instruments”
The nodes inside the instrument folders represent “detectors.”
One instrument can be associated with many different detectors. Each detector can have unique performance characteristics.
Detector - Si(Li) - Apr 12, 2012
Detector - Bruker-SLOW-5eV
JXA-8500F
Detector - Bruker 5 eV/ch
Probe
MIRA-3
Detector - PulseTor (Best)
FEI FIB
FPS
McCrone
Detector - EDAX
Detector - EDAX 4000
842.10 JEOL
Oxford
Detector - 80 mm²
Status
Enable detector
Import
Edit the properties of this detector.
OK
Cancel
Apply
Detector type
Silicon Drift Detector
Detector = Bruker 5 eV/ch - FWHM[Mn Kα]=130.1 eV - 2012-06-18 12:00

But I don’t know some of the values you ask about my detector…

Don’t fret. It is likely the vendor who sold you your detector doesn’t have good answers for every question. Even if they did, they don’t feel compelled to tell us. Some claim that it would put them at a competitive disadvantage if they told us. More really don’t know. The vendor that they purchase there detector crystals won’t tell them.

Fortunately, for almost every purpose, it really doesn’t matter.
(This seems to bother some OC people.)

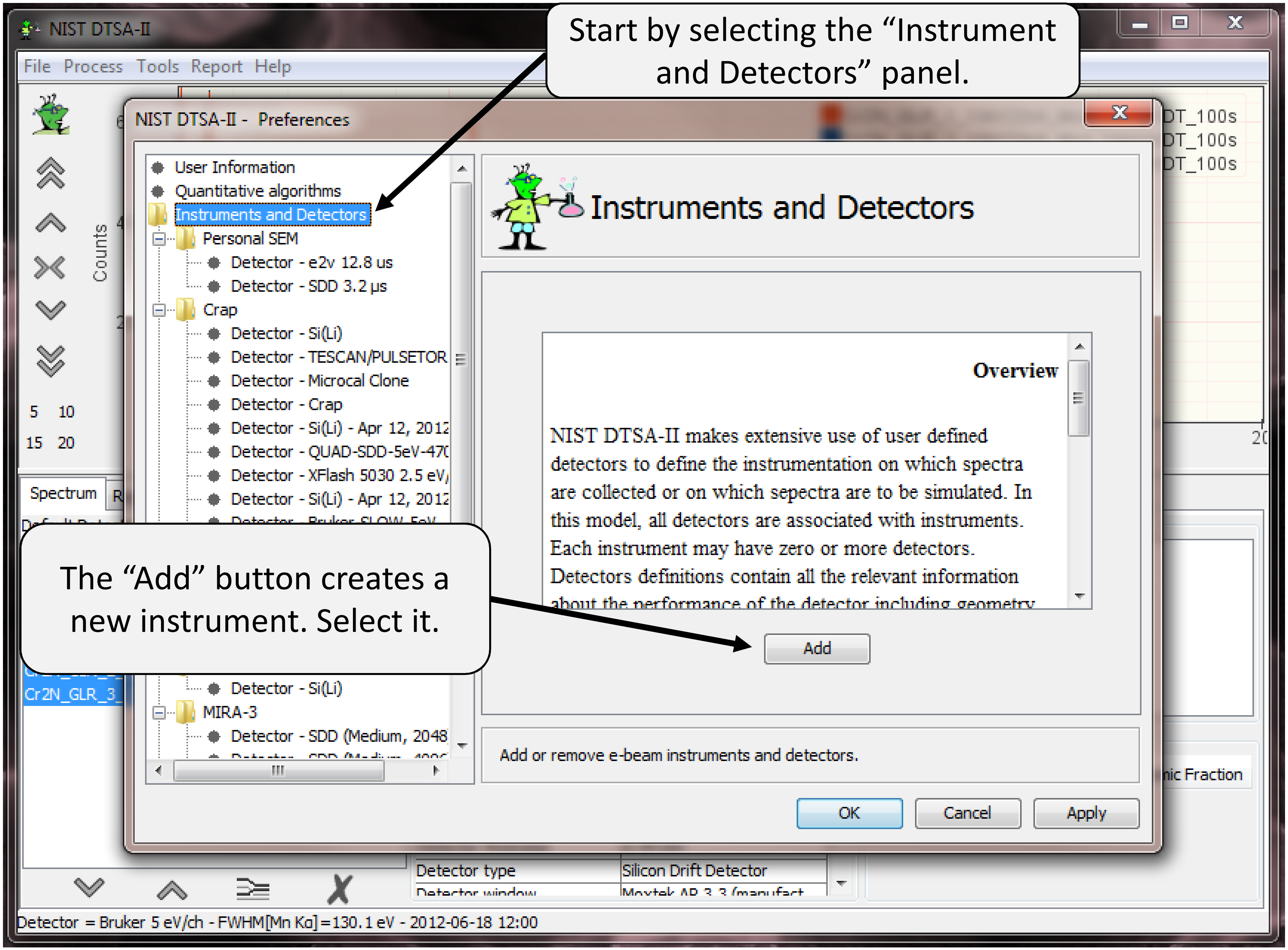
So why do I continue to ask? Maybe your purpose is special and it does matter. If this is the case (and you will know), then DTSA-II will allow you to specify the details.

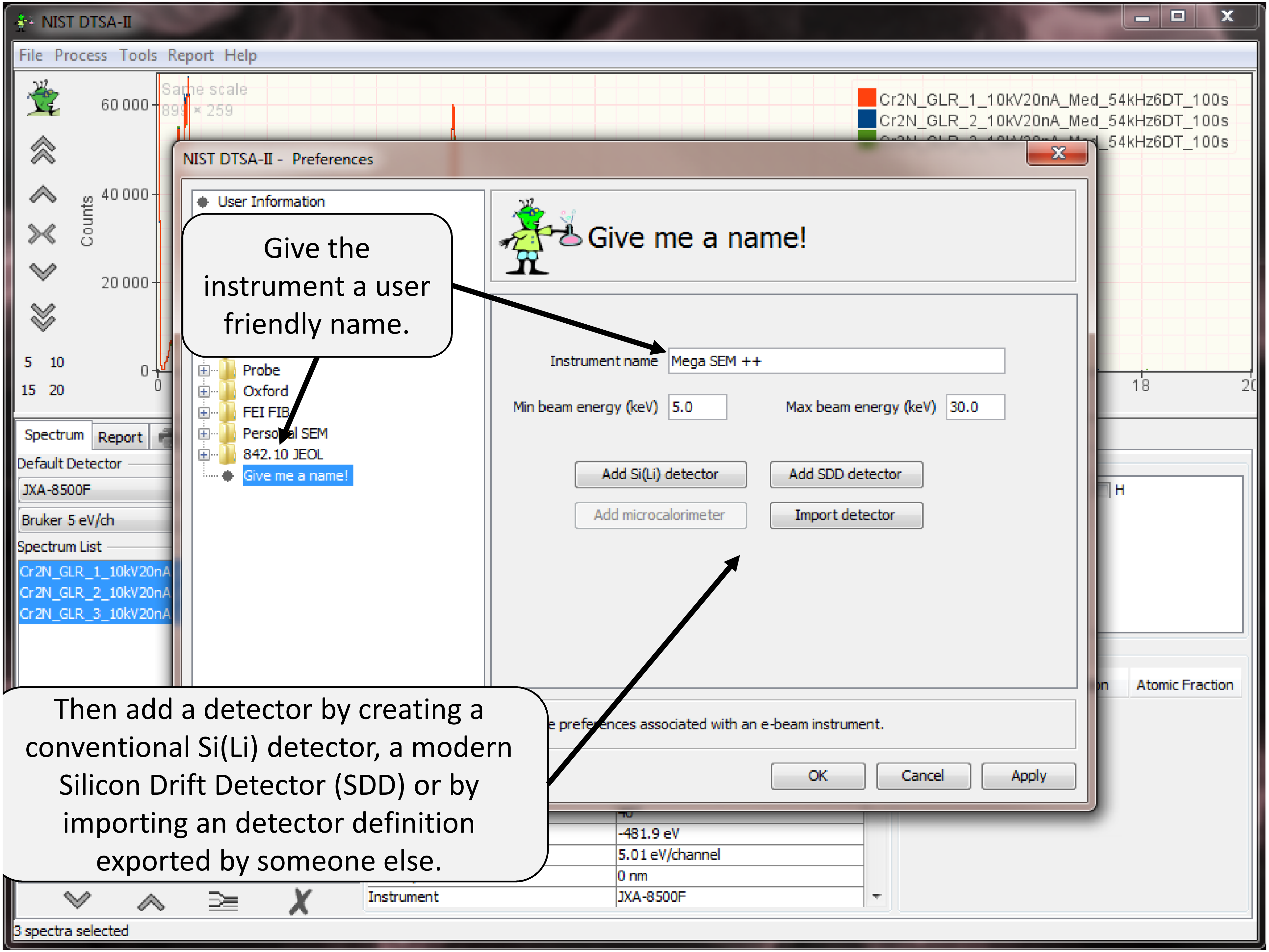
Personally, I don’t sweat it too much. Usually, I take most of the default values unless I know otherwise. Some parameters do matter and…

I will point out those parameters that do matter.

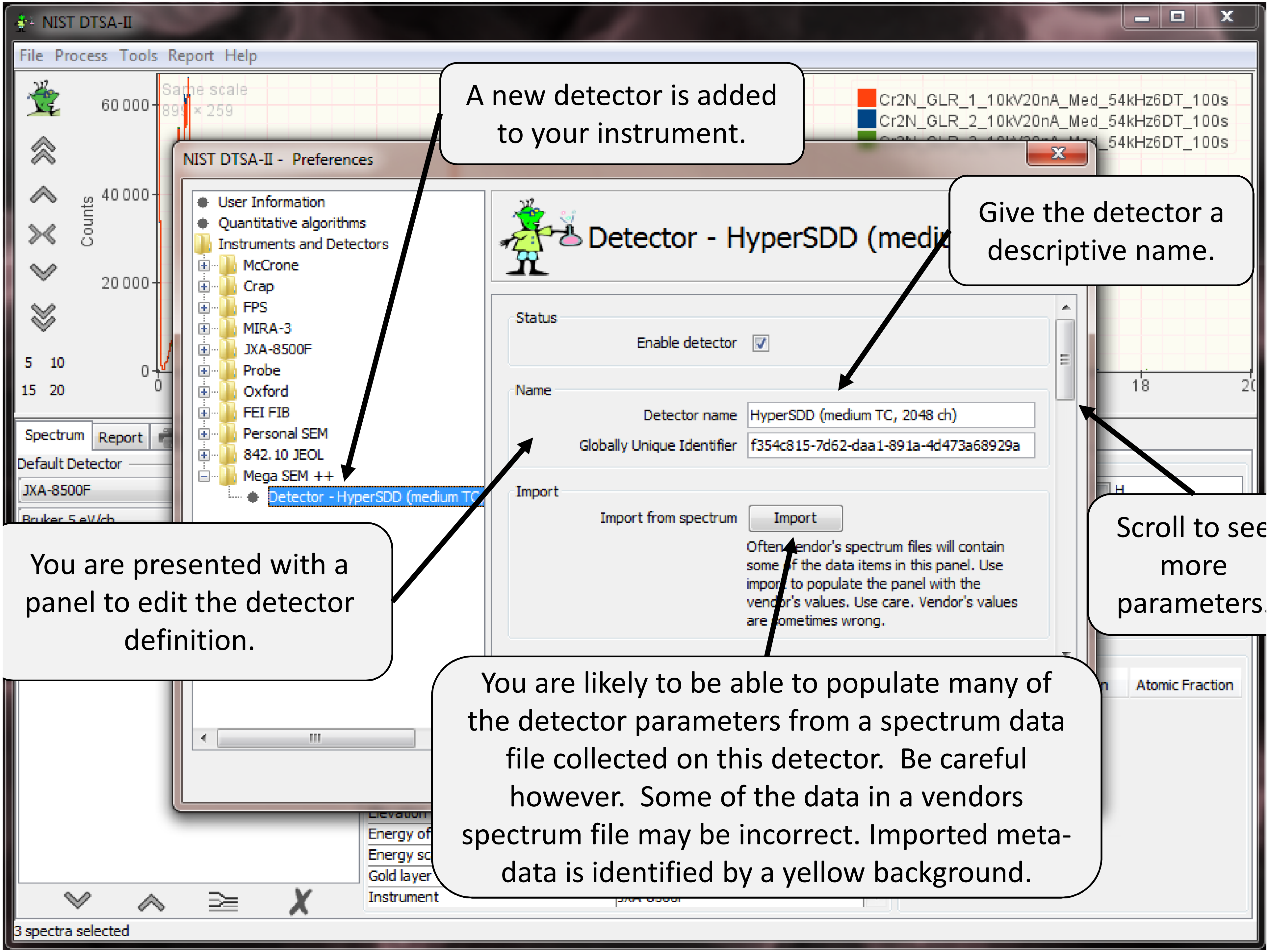
Start by selecting the “Instrument and Detectors” panel.

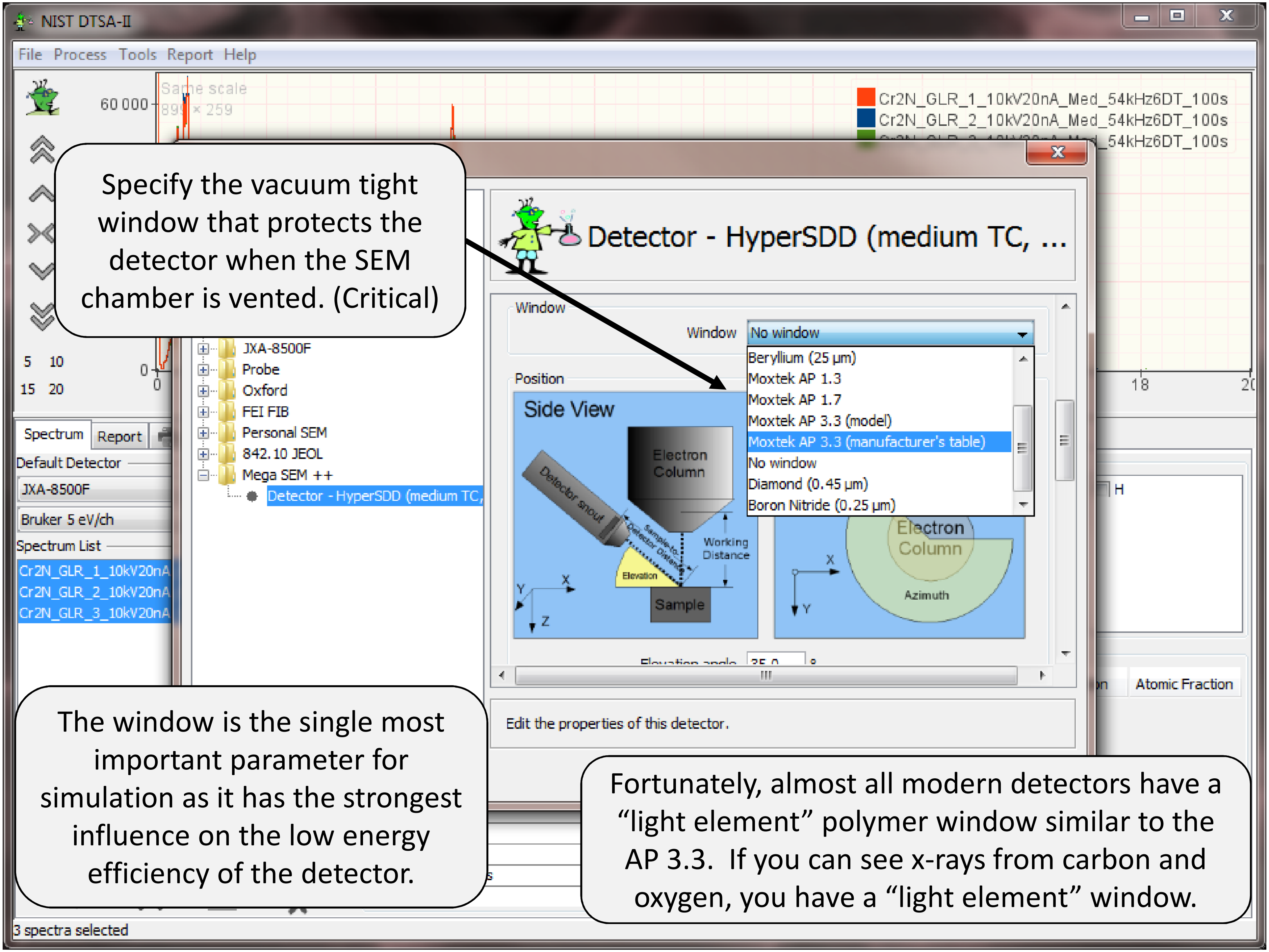
The “Add” button creates a new instrument. Select it.





NIST DTSA-II
File Process Tools Report Help
Same scale
Counts
60 000
40 000
20 000
0
Cr2N_GLR_1_10kV20nA_Med_54kHz6DT_100s
Cr2N_GLR_2_10kV20nA_Med_54kHz6DT_100s
Spectrum
Report
Default Detector
JXA-8500F
Bruker 5 eV/ch
Spectrum List
Cr2N_GLR_1_10kV20nA
Cr2N_GLR_2_10kV20nA
Cr2N_GLR_3_10kV20nA
NIST DTSA-II - Preferences
User Information
Probe
Oxford
FEI FIB
Personal SEM
842.10 JEOL
Give me a name!
Give me a name!
Instrument name
Mega SEM ++
Min beam energy (keV)
5.0
Max beam energy (keV)
30.0
Add Si(Li) detector
Add SDD detector
Add microcalorimeter
Import detector
e preferences associated with an e-beam instrument.
OK
Cancel
Apply
Atomic Fraction
-481.9 eV
5.01 eV/channel
0 nm
Instrument
JXA-8500F
3 spectra selected
Give the instrument a user friendly name.
Then add a detector by creating a conventional Si(Li) detector, a modern Silicon Drift Detector (SDD) or by importing an detector definition exported by someone else.


A new detector is added to your instrument.
Give the detector a descriptive name.
Scroll to see more parameters.
You are presented with a panel to edit the detector definition.
You are likely to be able to populate many of the detector parameters from a spectrum data file collected on this detector. Be careful however. Some of the data in a vendors spectrum file may be incorrect. Imported meta-data is identified by a yellow background.
NIST DTSA-II
File Process Tools Report Help
NIST DTSA-II - Preferences
User Information
Quantitative algorithms
Instruments and Detectors
McCrone
Crap
FPS
MIRA-3
JXA-8500F
Probe
Oxford
FEI FIB
Personal SEM
842.10 JEOL
Mega SEM ++
Detector - HyperSDD (medium TC
Status
Enable detector
Name
Detector name
HyperSDD (medium TC, 2048 ch)
Globally Unique Identifier
f354c815-7d62-daa1-891a-4d473a68929a
Import
Import from spectrum
Import
Spectrum
Report
Default Detector
JXA-8500F
Atomic Fraction
3 spectra selected

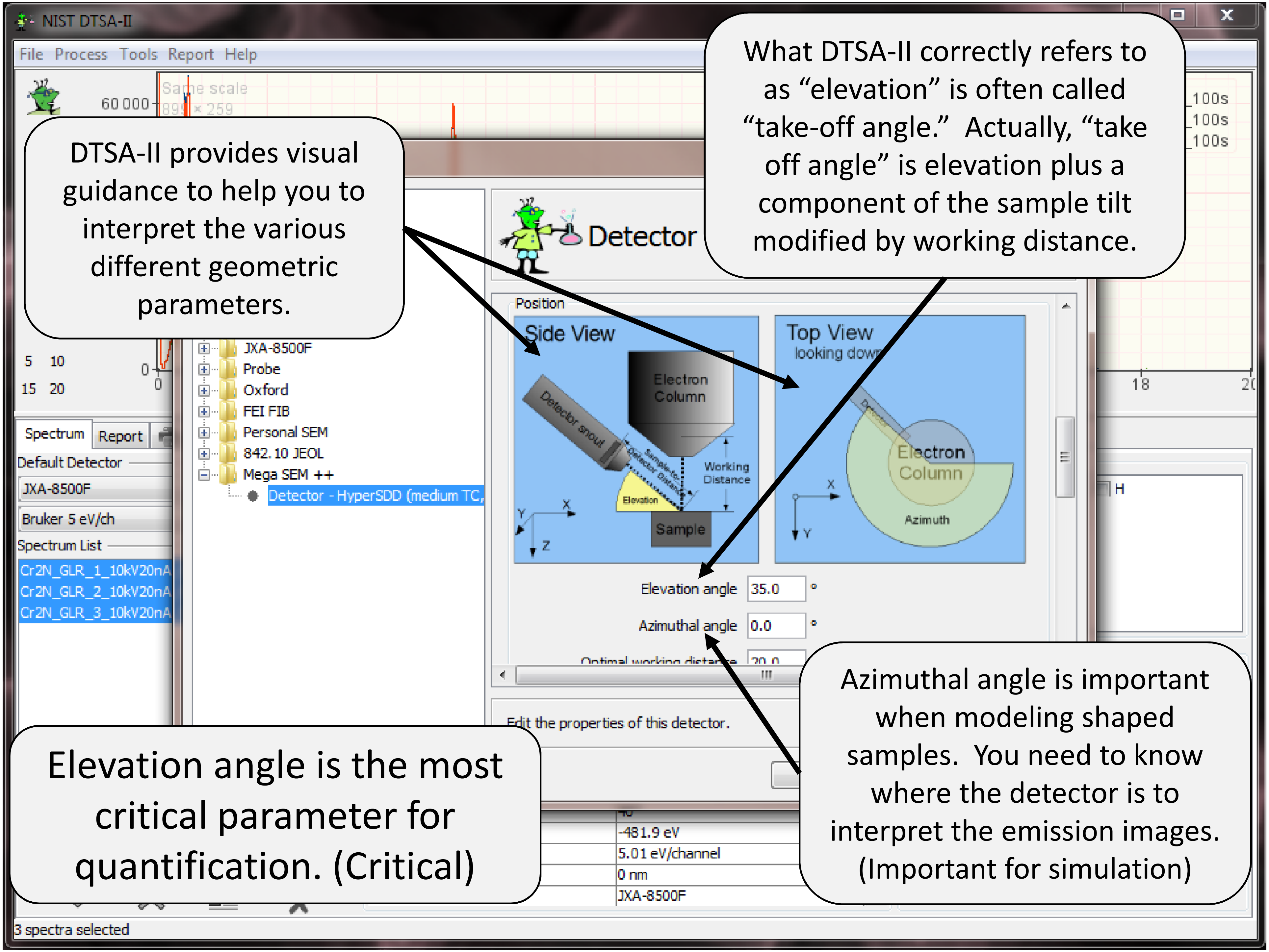

NIST DTSA-II
File Process Tools Report Help
Specify the vacuum tight window that protects the detector when the SEM chamber is vented. (Critical)
Detector - HyperSDD (medium TC, ...
Window
No window
Beryllium (25 µm)
Moxtek AP 1.3
Moxtek AP 1.7
Moxtek AP 3.3 (model)
Moxtek AP 3.3 (manufacturer's table)
No window
Diamond (0.45 µm)
Boron Nitride (0.25 µm)
Position
Side View
Electron Column
Detector snout
Working Distance
Elevation
Sample
Azimuth
JXA-8500F
Probe
Oxford
FEI FIB
Personal SEM
842.10 JEOL
Mega SEM ++
Detector - HyperSDD (medium TC,
Spectrum
Report
Default Detector
Bruker 5 eV/ch
Spectrum List
Cr2N_GLR_1_10kV20nA_Med_54kHz6DT_100s
Cr2N_GLR_2_10kV20nA_Med_54kHz6DT_100s
Edit the properties of this detector.
Atomic Fraction
The window is the single most important parameter for simulation as it has the strongest influence on the low energy efficiency of the detector.
Fortunately, almost all modern detectors have a “light element” polymer window similar to the AP 3.3. If you can see x-rays from carbon and oxygen, you have a “light element” window.
3 spectra selected

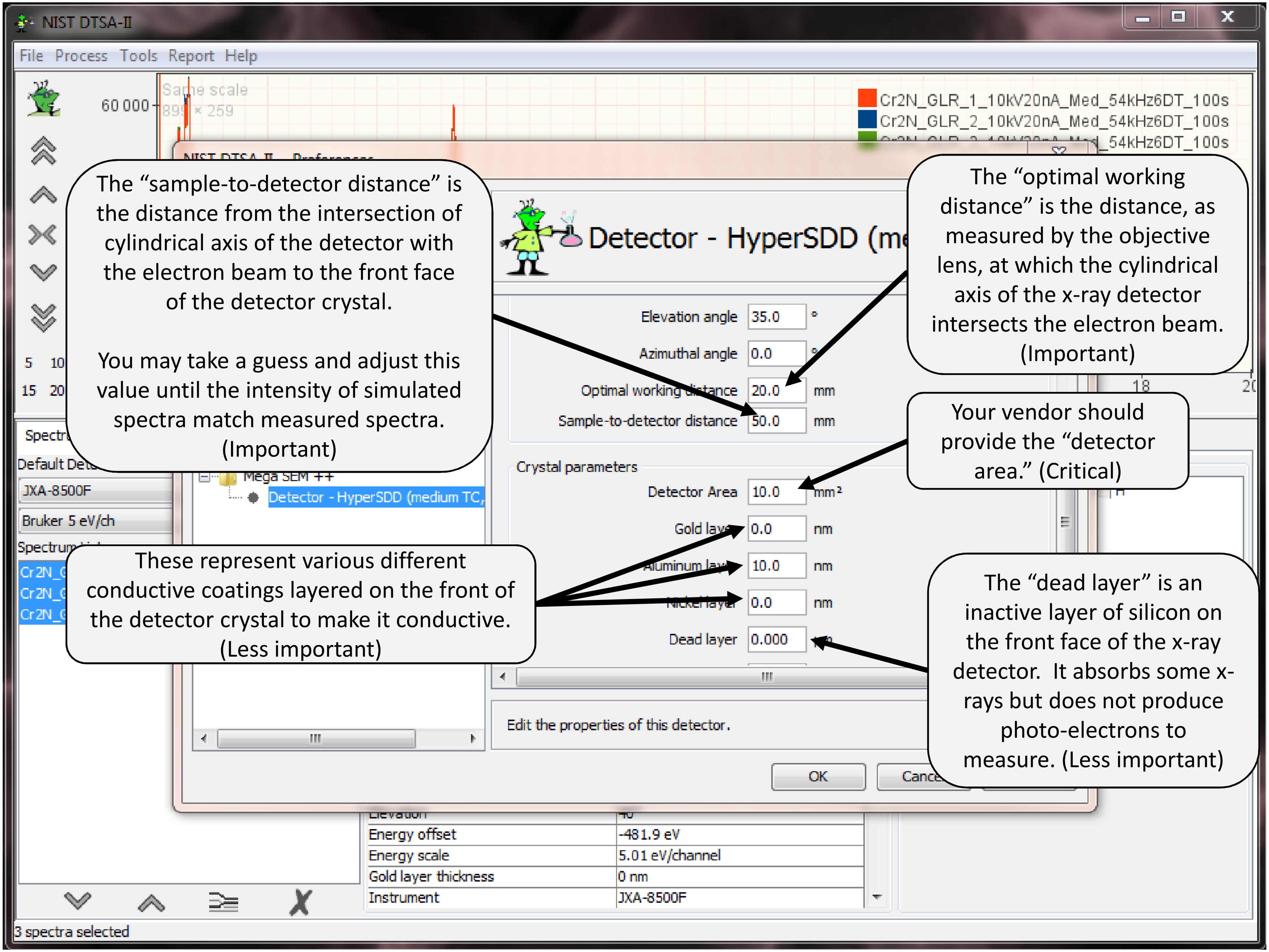
DTSA-II provides visual guidance to help you to interpret the various different geometric parameters.

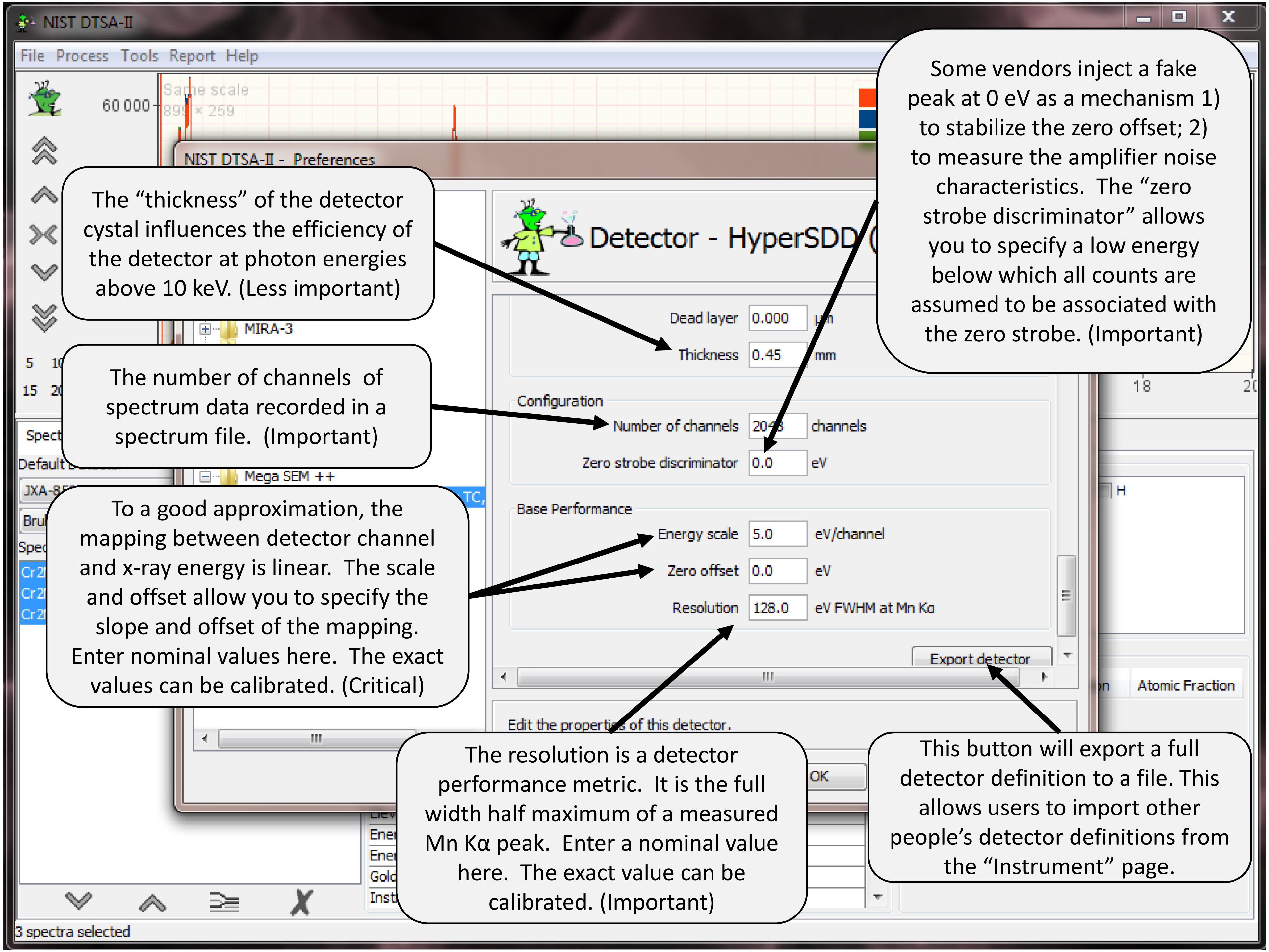
What DTSA-II correctly refers to as “elevation” is often called “take-off angle.” Actually, “take off angle” is elevation plus a component of the sample tilt modified by working distance.

Elevation angle is the most critical parameter for quantification. (Critical)

Azimuthal angle is important when modeling shaped samples. You need to know where the detector is to interpret the emission images. (Important for simulation)



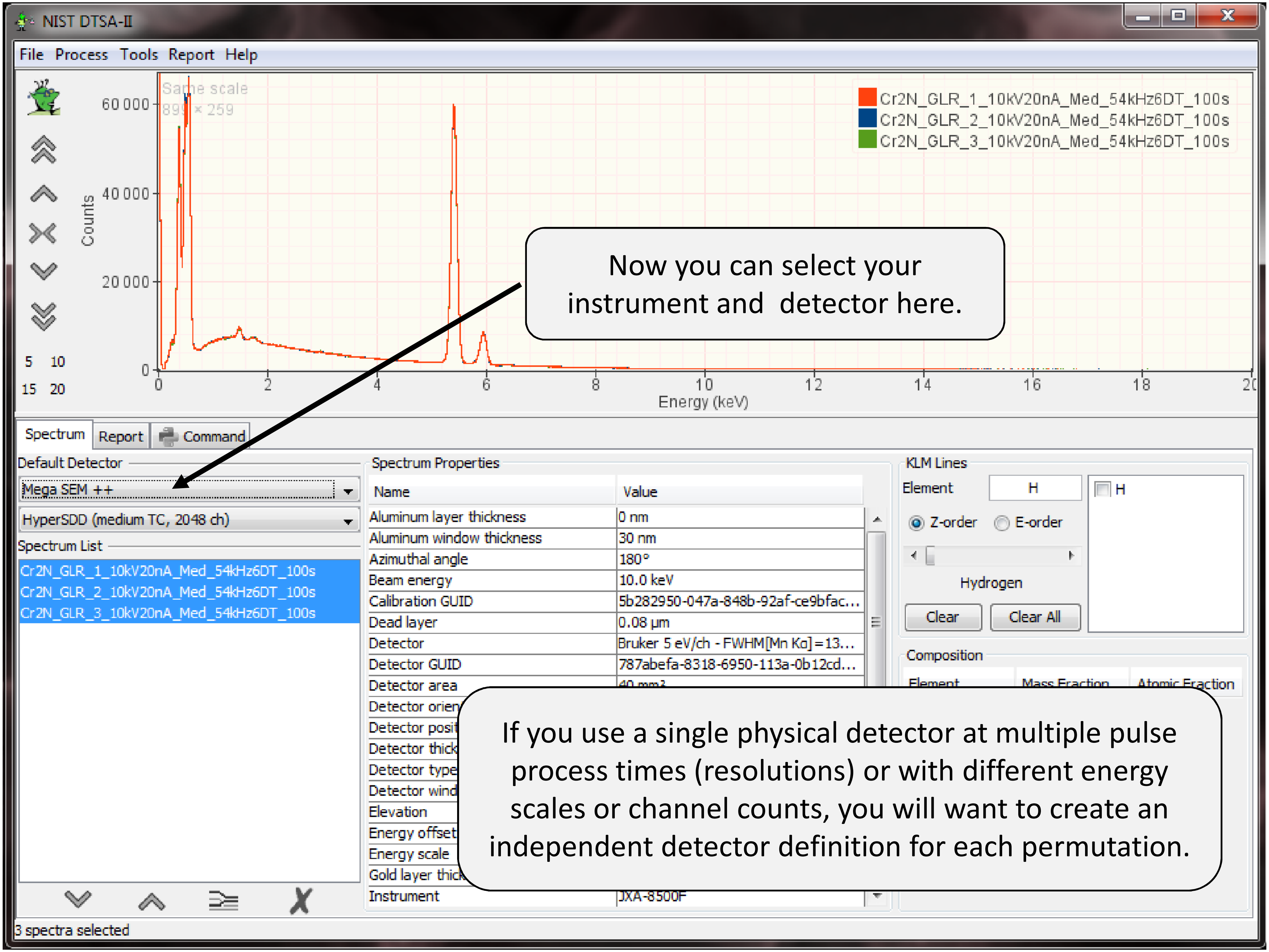

NIST DTSA-II
File Process Tools Report Help
Detector - HyperSDD (me
The “sample-to-detector distance” is the distance from the intersection of cylindrical axis of the detector with the electron beam to the front face of the detector crystal.
You may take a guess and adjust this value until the intensity of simulated spectra match measured spectra. (Important)
The “optimal working distance” is the distance, as measured by the objective lens, at which the cylindrical axis of the x-ray detector intersects the electron beam. (Important)
Elevation angle 35.0 °
Azimuthal angle 0.0 °
Optimal working distance 20.0 mm
Sample-to-detector distance 50.0 mm
Crystal parameters
Detector Area 10.0 mm²
Gold layer 0.0 nm
Aluminum layer 10.0 nm
Nickel layer 0.0 nm
Dead layer 0.000
Your vendor should provide the “detector area.” (Critical)
These represent various different conductive coatings layered on the front of the detector crystal to make it conductive. (Less important)
The “dead layer” is an inactive layer of silicon on the front face of the x-ray detector. It absorbs some x-rays but does not produce photo-electrons to measure. (Less important)
Edit the properties of this detector.
OK
Detector - HyperSDD (medium TC,
JXA-8500F
Bruker 5 eV/ch
Energy offset -481.9 eV
Energy scale 5.01 eV/channel
Gold layer thickness 0 nm
Instrument JXA-8500F
3 spectra selected

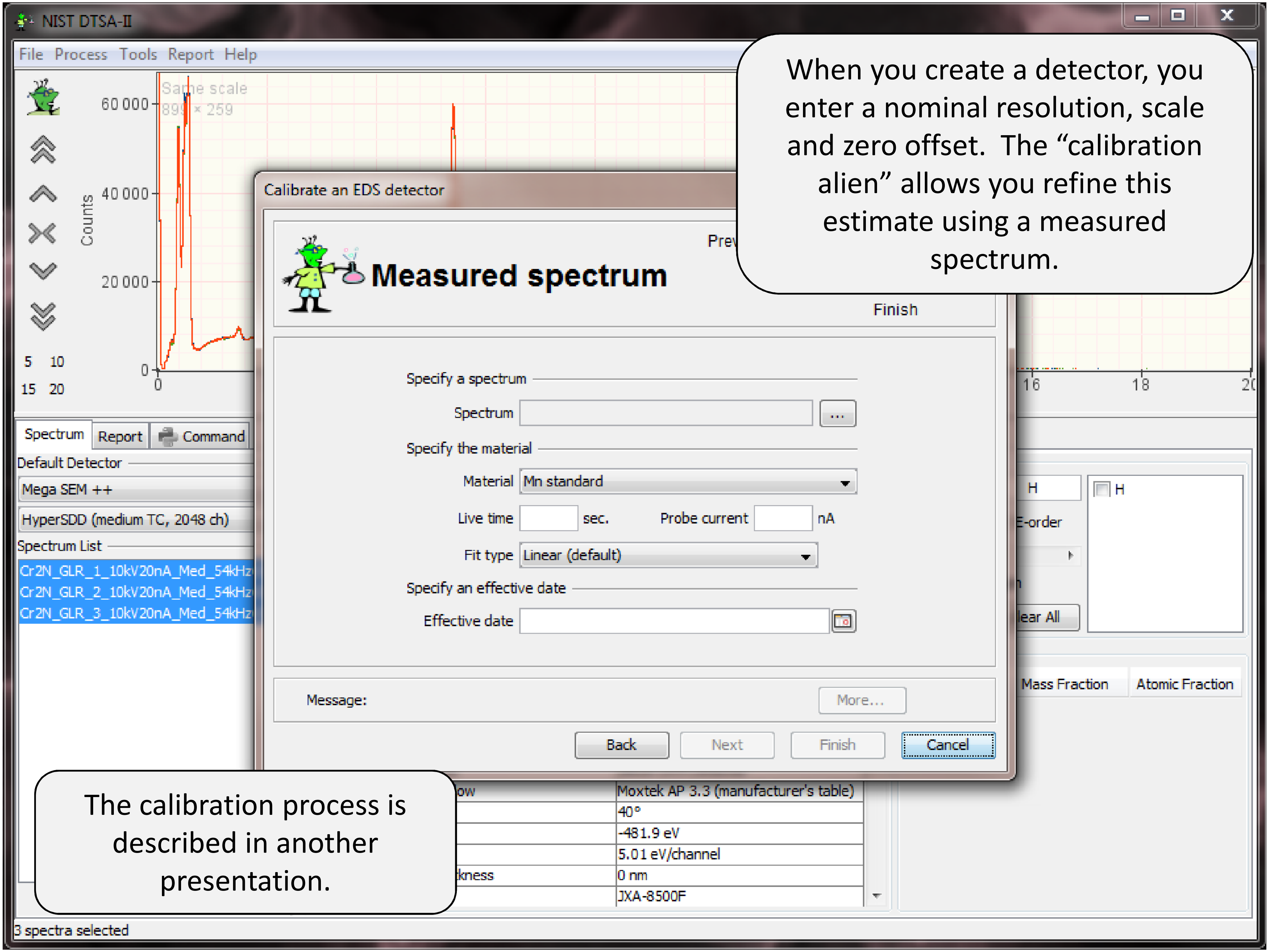

NIST DTSA-II
File Process Tools Report Help
NIST DTSA-II - Preferences
Detector - HyperSDD (
Dead layer 0.000 μm
Thickness 0.45 mm
Configuration
Number of channels 2048 channels
Zero strobe discriminator 0.0 eV
Base Performance
Energy scale 5.0 eV/channel
Zero offset 0.0 eV
Resolution 128.0 eV FWHM at Mn Kα
Export detector
Edit the properties of this detector.
OK
MIRA-3
Mega SEM ++
Atomic Fraction
3 spectra selected
The “thickness” of the detector cystal influences the efficiency of the detector at photon energies above 10 keV. (Less important)
The number of channels of spectrum data recorded in a spectrum file. (Important)
To a good approximation, the mapping between detector channel and x-ray energy is linear. The scale and offset allow you to specify the slope and offset of the mapping. Enter nominal values here. The exact values can be calibrated. (Critical)
Some vendors inject a fake peak at 0 eV as a mechanism 1) to stabilize the zero offset; 2) to measure the amplifier noise characteristics. The “zero strobe discriminator” allows you to specify a low energy below which all counts are assumed to be associated with the zero strobe. (Important)
The resolution is a detector performance metric. It is the full width half maximum of a measured Mn Kα peak. Enter a nominal value here. The exact value can be calibrated. (Important)
This button will export a full detector definition to a file. This allows users to import other people’s detector definitions from the “Instrument” page.

Parameters by importance

- Critical
  - “Window”, “Elevation angle”, “Detector area”, “Energy scale”, “Zero offset”
- Important
  - “Azimuthal angle”, “Optimal working distance”, “Sample-to-detector distance”, “Number of channels”, “Zero strobe discriminator”, “Resolution”
- Less important
  - “Gold layer”, “Nickel layer”, “Aluminum layer”, “Dead layer”, “Thickness”

Do your best. It will probably be good enough.


NIST DTSA-II
File Process Tools Report Help
Same scale
Counts
Energy (keV)
Cr2N_GLR_1_10kV20nA_Med_54kHz6DT_100s
Cr2N_GLR_2_10kV20nA_Med_54kHz6DT_100s
Cr2N_GLR_3_10kV20nA_Med_54kHz6DT_100s
Now you can select your instrument and detector here.
Spectrum
Report
Command
Default Detector
Mega SEM ++
HyperSDD (medium TC, 2048 ch)
Spectrum List
Spectrum Properties
Name
Value
Aluminum layer thickness
0 nm
Aluminum window thickness
30 nm
Azimuthal angle
180°
Beam energy
10.0 keV
Calibration GUID
5b282950-047a-848b-92af-ce9bfac...
Dead layer
0.08 µm
Detector
Bruker 5 eV/ch - FWHM[Mn Kα]=13...
Detector GUID
787abefa-8318-6950-113a-0b12cd...
Detector area
Elevation
Energy scale
Instrument
JXA-8500F
KLM Lines
Element
H
Z-order
E-order
Hydrogen
Clear
Clear All
Composition
If you use a single physical detector at multiple pulse process times (resolutions) or with different energy scales or channel counts, you will want to create an independent detector definition for each permutation.
3 spectra selected



When you create a detector, you enter a nominal resolution, scale and zero offset. The "calibration alien" allows you refine this estimate using a measured spectrum.

The calibration process is described in another presentation.